

Arjun Giriraj Day 06

1. What is IPv4?

IPv4 is a logical addressing for devices communicates over IP network. An IPv4 contains of 32 bit of address written as 4 decimal numbers separated by dots.

2. How many bits are in an IPv4 address?

32 bits.

3. What is an octet?

The four groups in IPv4 address is called octets. Each octet is of 8 bit so $8+8+8+8$ is 32 bits.

4. How many octets are in an IPv4 address?

4 octets.

5. What is a subnet mask?

It tells the device which portion of address is network.

6. Difference between network portion and host portion?

IPv4 address is separated in two parts network portion and host portion, first 3 octets is network portion and last octet is host portion.

7. What is a private IPv4 address?

Private IPv4 address is not routable on the public internet. It is used in homes, offices, schools, etc.

8. Write all three private IPv4 ranges.

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

9. What is a public IPv4 address?

Public IPv4 is globally routable address used for internet communication.

10. What is the IPv4 loopback address?

127.0.0.0/8

11. What is APIPA?

APIPA is an IPv4 address which windows self assign to itself when there is any problem in DHCP or network configuration.

12. What does 169.254.x.x commonly indicate?

DHCP/network configuration problem.

13. What is a default gateway?

Default gateway is to where the router sends packet to send outside the network.

14. Static vs dynamic IP addressing?

Static IP addressing is manually configured and kept consistent, it is used for servers, printers, etc.

Dynamic IP addressing is assigned automatically by DHCP(Dynamic Host Configuration Protocol).

15. What is DHCP?

Dynamic Host Configuration Protocol, it is a protocol which assigns devices IP address, Subnet masking, Gateway, and DNS.

16. What is a network address?

Network address identifies the network itself.

17. What is a broadcast address?

Broadcast address is a network address used to transmit to all devices connected in same network.

18. Why is IPv4 knowledge important in cybersecurity?

Because Analysts works with ip address, server logs, Packet captures, firewall logs, nmap result and wireshark traffic.

Your own: IPv4 address, Subnet mask, Default gateway, DNS server, DHCP status.

IPv4 address: 10.230.76.85

Subnet Mask: 255.255.255.0

Default gateway: 10.230.76.191

DNS server: 10.230.76.191

DHCP status: auto

Attach screenshots of:

ipconfig

```
venom@VENOM:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4917 bytes 536057 (523.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4917 bytes 536057 (523.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.230.76.85 netmask 255.255.255.0 broadcast 10.230.76.255
    inet6 2401:4900:46a7:c7c4:9671:73f7:1469:3c09 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::83a4:13e:861d:1ec prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:bc61:f58d:bde3:22c1:e232:a4ab prefixlen 64 scopeid 0x0<global>
    ether 10:0f:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 348322 bytes 435884405 (415.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 179753 bytes 28438598 (27.1 MiB)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0
```

ipconfig /all

```
venom@VENOM:~$ ifconfig -a
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4917 bytes 536057 (523.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4917 bytes 536057 (523.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.230.76.85 netmask 255.255.255.0 broadcast 10.230.76.255
    inet6 2401:4900:46a7:c7c4:9671:73f7:1469:3c09 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::83a4:13e:861d:1ec prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:bc61:f58d:bde3:22c1:e232:a4ab prefixlen 64 scopeid 0x0<global>
    ether 10:0f:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 348326 bytes 435884785 (415.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 179758 bytes 28439260 (27.1 MiB)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0
```

ping 127.0.0.1

```
venom@VENOM:~$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.032 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.027 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.023 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.028 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.031 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.031 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.024 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.027 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.023 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.028 ms
^C
--- 127.0.0.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9187ms
rtt min/avg/max/mdev = 0.023/0.027/0.032/0.003 ms
```